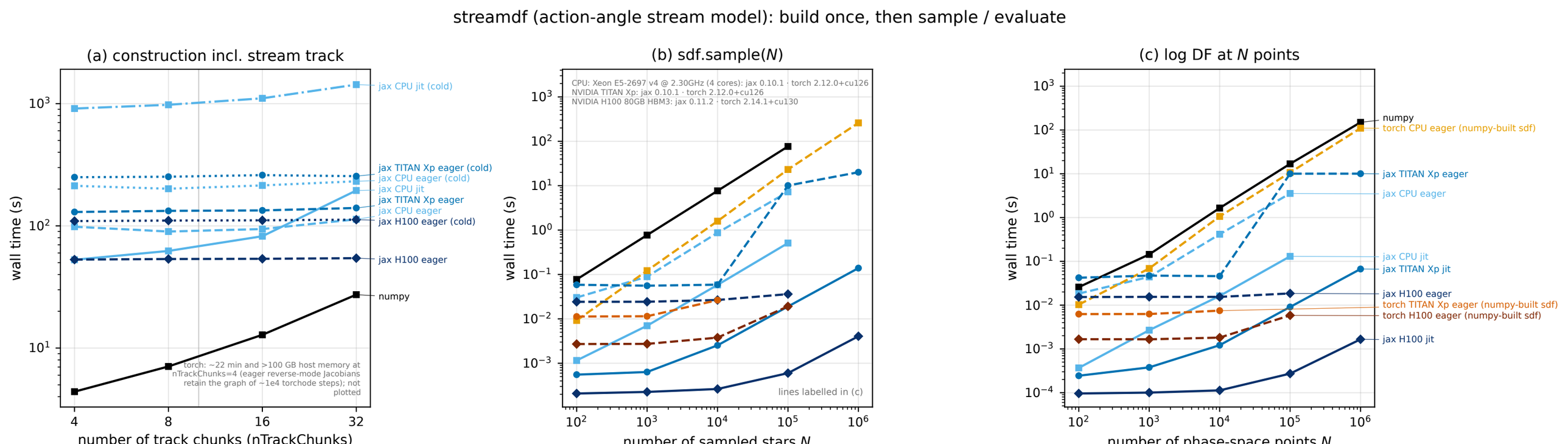
streamdf (action-angle stream model): build once, then sample / evaluate
(a) construction incl. stream track
wall time (s)
number of track chunks (nTrackChunks)
jax CPU jit (cold)
jax TITAN Xp eager (cold)
jax CPU eager (cold)
jax CPU jit
jax TITAN Xp eager
jax CPU eager
jax H100 eager (cold)
jax H100 eager
numpy
torch: ~22 min and >100 GB host memory at nTrackChunks=4 (eager reverse-mode Jacobians retain the graph of ~1e4 torchode steps); not plotted
(b) sdf.sample(N)
CPU: Xeon E5-2697 v4 @ 2.30GHz (4 cores): jax 0.10.1 · torch 2.12.0+cu126
NVIDIA TITAN Xp: jax 0.10.1 · torch 2.12.0+cu126
NVIDIA H100 80GB HBM3: jax 0.11.2 · torch 2.14.1+cu130
wall time (s)
number of sampled stars N
lines labelled in (c)
(c) log DF at N points
wall time (s)
number of phase-space points N
numpy
torch CPU eager (numpy-built sdf)
jax TITAN Xp eager
jax CPU eager
jax CPU jit
jax TITAN Xp jit
jax H100 eager
torch TITAN Xp eager (numpy-built sdf)
torch H100 eager (numpy-built sdf)
jax H100 jit